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B.Tech/M.Tech(Integrated) DEGREE EXAMINATION, DECEMBER 2023

Fifth Semester

21CSC302J - COMPUTER NETWORKS

(For the candidates admitted during the academic year 2022-2023 onwards)

Note:

i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours **Max. Marks: 75**

PART - A (20 × 1 = 20 Marks)

Answer **all** Questions

		Marks	BL	CO
1.	The IP checksum must be recalculated on each router due to the modification of the _____ fields. (A) TTL, Options, Identifier, Offset (B) TTL, Options, Datagram length, Offset (C) TTL, Options, Data, Offset (D) TTL, Header length, Offset, ToS (Type of service)	1	1	1
2.	When the source does not trust the routers to route correctly, what options can be used? (A) Loose source routing (B) Trace route (C) Strict source routing (D) Internet Time Stamp	1	2	1
3.	Suppose a router receives an IP packet containing 600 bytes of data and forwards it to a network with an MTU of 200 bytes. Assume that the IP header is 20 bytes long. What are the fragment offset values for fragmented packets? (A) 22, 44, 66 (B) 0, 22, 44, 66 (C) 0, 22, 44 (D) 22, 44, 66, 88	1	3	1
4.	Which fields are required in the IP header to allow the destination to perform fragment reassembly? (A) Identifier, MF (More Fragments), Offset, Header Length (IHL) and Total Length (B) MF (More Fragments), Offset and destination IP address (C) MF (More Fragments), datagram length, and destination IP address (D) MF (More Fragments), options and offset	1	5	1
5.	The _____ high order byte is stored on starting address and low order byte is stored on next address. (A) Big- endian (B) Little-endian (C) System calls (D) Byte ordering	1	1	2
6.	The _____ function converts 32-bits (4-byte) data from host byte order to network byte order. (A) unsigned shot htons() (B) unsigned shot htonl() (C) unsigned shot ntohs() (D) unsigned shot htohl()	1	1	2
7.	Stream control Transmission Protocol (SCTP) is a _____ Protocol. (A) Unreliable, character- oriented (B) Reliable, message- oriented (C) Unreliable, message- oriented (D) Reliable, character- oriented	1	2	2

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8. Most application on the internet are designed using a client-server paradigm in which an _____ program, called a server, provides services and another _____ program, called client, receives services. 1 3 2

(A) Application (B) Server
(C) Client (D) Systems

9. The right to use a domain name is delegated by domain name registers which are accredited by _____. 1 1 3

(A) internet architecture board (B) internet society
(C) internet research task force (D) internet corporation for assigned names and numbers

10. If 5 files are transferred from server A to client B in the same session. The number of TCP connections between A and B is _____. 1 4 3

(A) 5 (B) 10
(C) 2 (D) 6

11. The HTTP response message leaves out the requested object when _____ method is used. 1 2 3

(A) GET (B) POST
(C) HEAD (D) PUT

12. Match the following. 1 3 3

(P) SMTP	(1) Application layer
(Q) BGP	(2) Transport layer
(R) TCP	(3) Data link layer
(S) PPP	(4) Network layer
	(5) Physical layer

(A) P – 2; Q – 1; R – 3; S – 5 (B) P – 1; Q – 4; R – 2; S – 3
(C) P – 1; Q – 4; R – 2; S – 5 (D) P – 2; Q – 4; R – 1; S – 3

13. Suppose two IPv6 nodes want to interoperate using IPv6 datagrams, but they are connected to each other by intervening IPv4 routers. The best solution here is _____. 1 1 4

(A) Use dual-stack approach (B) Tunneling
(C) No solution (D) Replace the system

14. Teredo is an automatic tunneling technique. In each client the obfuscated IPv4 address is represented by bits _____. 1 2 4

(A) 96 to 127 (B) 0 to 63
(C) 80 to 95 (D) 64 to 79

15. The _____ are used for communication among IPv6 hosts on a link (broadcast segment) only 1 3 4

(A) Global unicast address (B) Unique local address
(C) Local link address (D) Address between Global and Local link address

16. Which of the following is the class of range A public IPV4? 1 2 4

(A) 0.0.0.0 to 233.255.255.255 (B) 0.0.0.0 to 192.0.0.0
(C) 0.0.0.0 to 127.255.255.255. (D) 0.0.0.0 to 128.0.0.0

17. Which protocol does the PPP protocol provide for handling the capabilities of the connection/link on the network? 1 1 5

(A) LCP (B) NCP
(C) Both LCP and NCP (D) TCP

18. The link necessarily begins and ends with this phase. During the _____ phase, the LCP automata will be in INITIAL or STARTING states. 1 2 5

(A) Link-termination phase (B) Link establishment phase
(C) Authentication phase (D) Link dead phase

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19. Virtual Private Network (VPN) is one of the applications of _____. 1 2 5

(A) MAC Protocols (B) SMTP
(C) IPSec (D) TLS Protocol

20. Which of the following is the shortest valid abbreviation for DE80:0000:0000:0100:0000:0000:0000:0123? 1 3 5

(A) DE80::100::123 (B) DE8::1::123
(C) DE80::100:0:0:0:123 (D) DE80:0:0:100::1230

PART - B (5 × 8 = 40 Marks) Marks BL CO

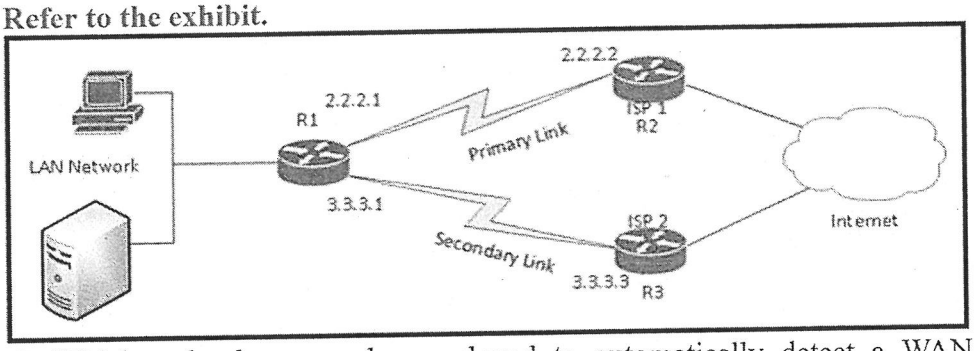
Answer **all** Questions

21. (a) (i) An IP packet has arrived in which the offset value is 100, the value of HLEN is 5 and the value of the total length field is 100. What is the number of the first byte and the last byte? (5) 8 2 1
(ii) List out the components of ARP packages and How the cache-control module is responsible for maintaining the cache table. (3)

(OR)

(b) (i) Draw the ICMP header with proper values for "Source quench" type error reporting. (5)
(ii) Discuss about ICMP debugging tools.(3)

22. (a) **Refer to the exhibit.** 8 4 2



(i) Which technology can be employed to automatically detect a WAN primary link failure and failover to the secondary link? (2)

1. HSRP
2. VRRP
3. IP SLA
4. multicast

(ii) R1 is 3 router hops away from subnet 10.1.1.0/24. According to the show interfaces commands, all three links between R1 and 10.1.1.0/24 use the following settings: bandwidth: 1000, 500, 100000 and delay: 12000, 8000, 100. What delay and bandwidth value will be used to calculate the EIGRP metric? (6)

(OR)

(b) Write short notes on TCP Congestion Avoidance Phase.

23. (a) Assume you need to retrieve a scientific document that contains one reference to another text file and one reference to a large image. The main document and the image are stored in two separate files on the same site (file A and file B); the referenced text file is stored on another site (file C). Demonstrate the three transactions to see the whole document. Also, give the uniform resource locator format to locate any kind of information on the Internet. 8 3 3

(OR)

(b) Why Message Access Agent is required, and with a neat interaction diagram, specify the interaction between user computer and POP3 server

24. (a) Elaborate in brief about IPv6 routing protocols that enable routers to exchange information about connected networks. (Any 3 protocols) 8 1 4

(OR)

(b) Draw and explain the three levels of hierarchy of global unicast address.

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25.	(a) Sketch and discuss in detail about the ATM protocol architecture.	8	2	5
	(OR)			
	(b) Explain how VPN is designed to securely connect two geographically-distributed sites.			

PART - C (1 × 15 = 15 Marks)	**Marks**	**BL**	**CO**
Answer **any 1** Questions			
26. (i) Write short notes on Silly Window Syndrome in TCP flow control. (5) (ii) Calculate the checksum for the following ICMP (10) packet: Type: Echo Request Identifier : 123 Sequence number : 20 Message : COMPUTING	15	3	1
27. (i) DSL uses a modulation technique called DMT. Find some information about this modulation technique and how it can be used in DSL (8) (ii) PPP goes through different phases, which can be shown in a transition state diagram. Find the transition diagram for PPP connection (7)	15	3	4

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